

Shielded PPO Extension: Results

Muntasir Hossain (210041265). Thesis: Navigating Unsignalised Intersections in Frenet Space with PPO and Attention, CSE 4700, Islamic University of Technology. Scope: Section 11.2 of the pre-defence report.

Summary

1. A three-layer post-decision safety shield (path-based time-to-conflict braking, an RSS safe-distance override, right-before-left yielding) was implemented on the attention-based PPO controller, with every intervention counted per layer.
2. In the pre-defence environment, used for both training and evaluation, the original shield increased collisions from 6.35% to 9.72% ($p = 0.007$). Background vehicles there do not brake, so shield braking causes rear-end collisions. Combining rear-aware braking with a commit zone removed that harm: 6.35%, identical to the controller without a shield, and significantly safer than the original shield.
3. On the pre-defence benchmark, with the revised training pipeline, the shielded controller recorded 0.40% collisions against 0.88% reported for Attention + Continuous in the pre-defence study. Against identical weights without the shield (0.99%) the reduction is not statistically significant ($p = 0.18$).
4. The shield costs efficiency in every setting: longer travel time and a lower success rate, with the RSS layer producing most interventions.
5. Attention-based controllers significantly outperform heuristic controllers ($p < 0.001$), confirming the central pre-defence finding.
6. A first attempt at shield-aware training was stopped at 824k steps for the deadline and collided more often; with half the training budget this is inconclusive.

Conclusion

A post-decision shield did not yield a measurable net safety gain for this controller. Its original design harms safety in aggressive traffic; the rear-aware and commit-zone refinements remove that harm. Full-length shield-aware training against a matched baseline is the next step.

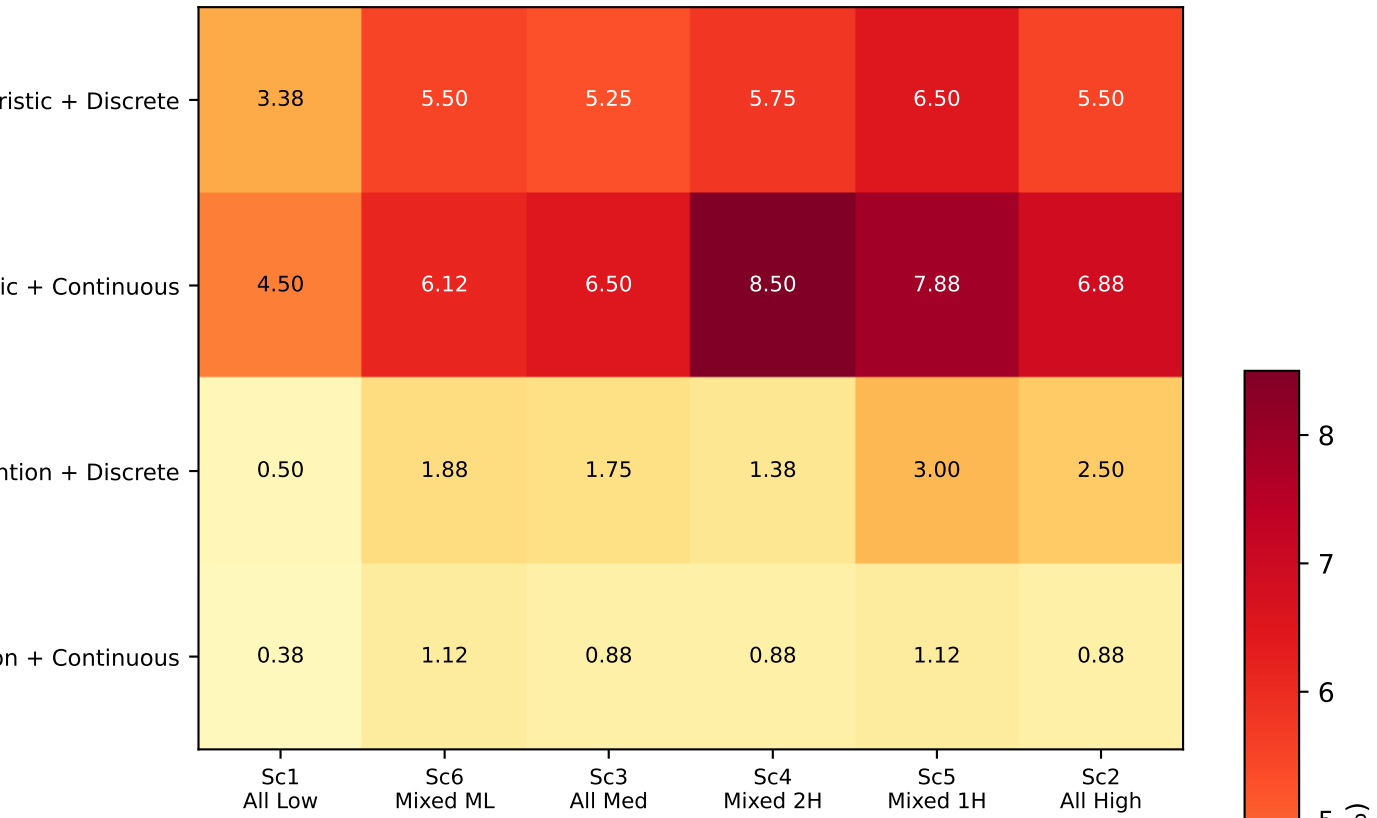
Evaluation and training environments

Setting	Pre-defence training	Revised pipeline training	Evaluation benchmark
Cross traffic	400 veh/h	275 veh/h	150 / 275 / 400 veh/h
Traffic in agent's lane	275 veh/h	none	none
Background speed_mode	0 (ignores safety)	31	31
RL spawn probability	0.3	0.8	0.8
Warmup	5 steps	50 steps	50 steps
Episodes	-	-	42 per configuration

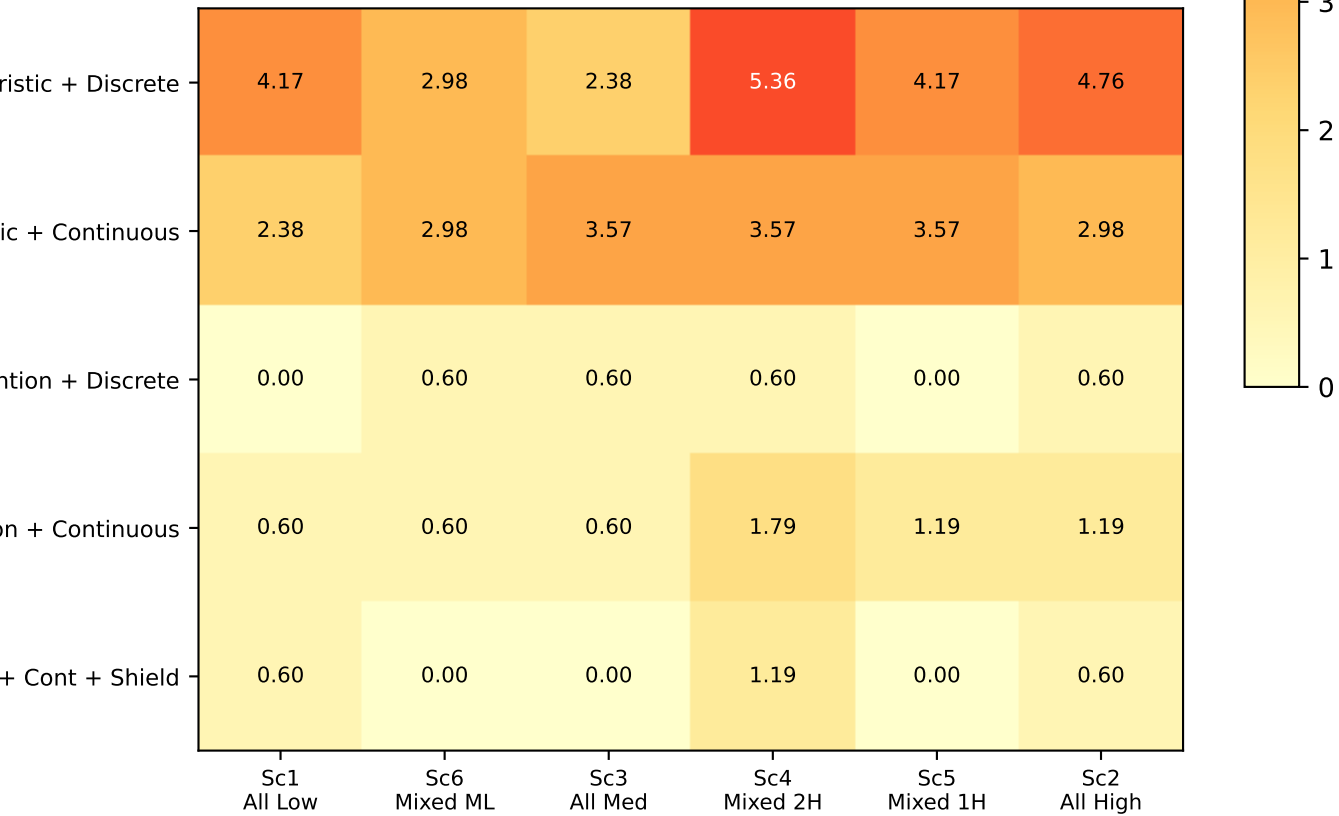
The evaluation benchmark is identical to the pre-defence protocol (`src/test/v0_1_evaluate.py`). Results labelled 'pre-defence environment' use the pre-defence training environment for both training and evaluation, reproduced from `src/configs/v0_1_single_agent.py`. The observation additionally includes three leader features added after the pre-defence report.

Collision rate by controller and traffic scenario

Pre-defence study (reported, Figure 9.2)

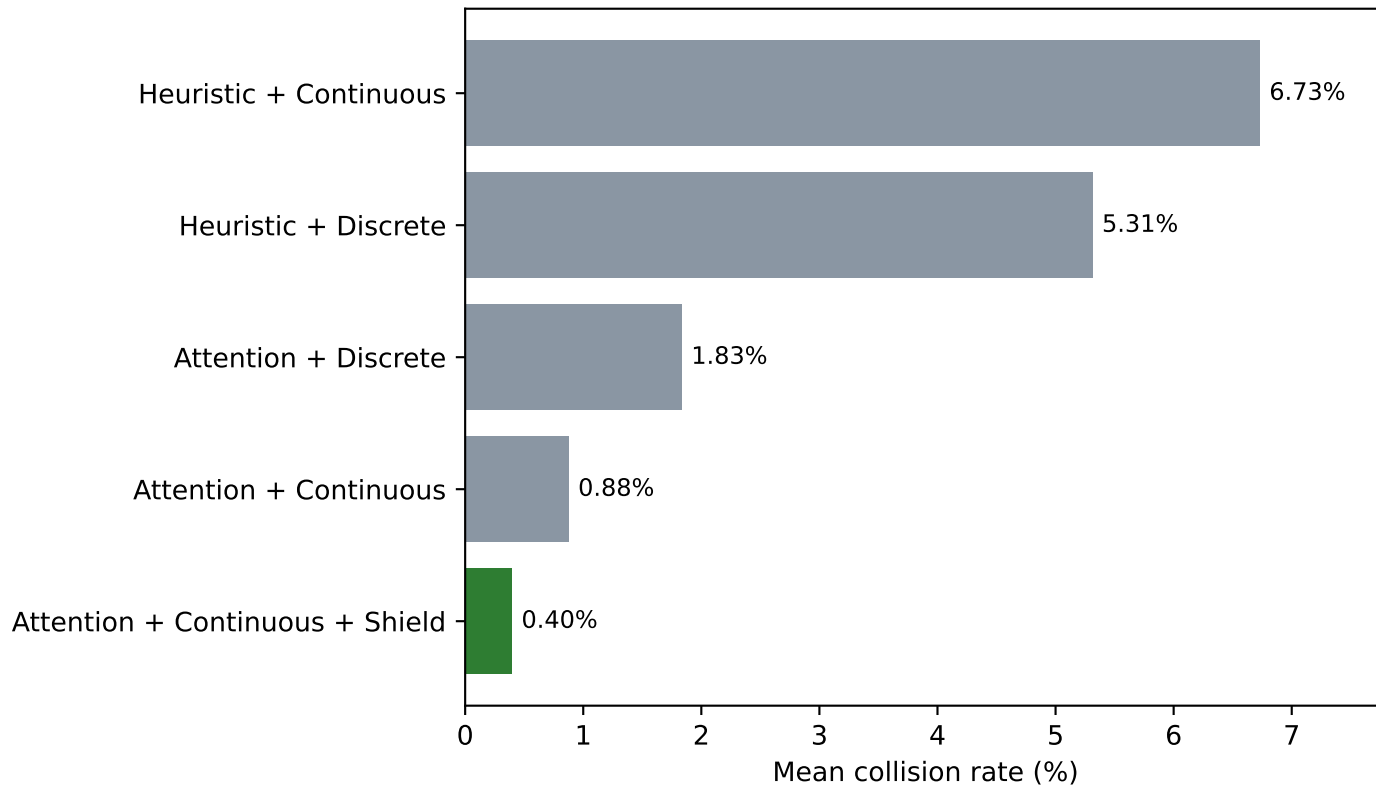


This work, revised pipeline, same benchmark



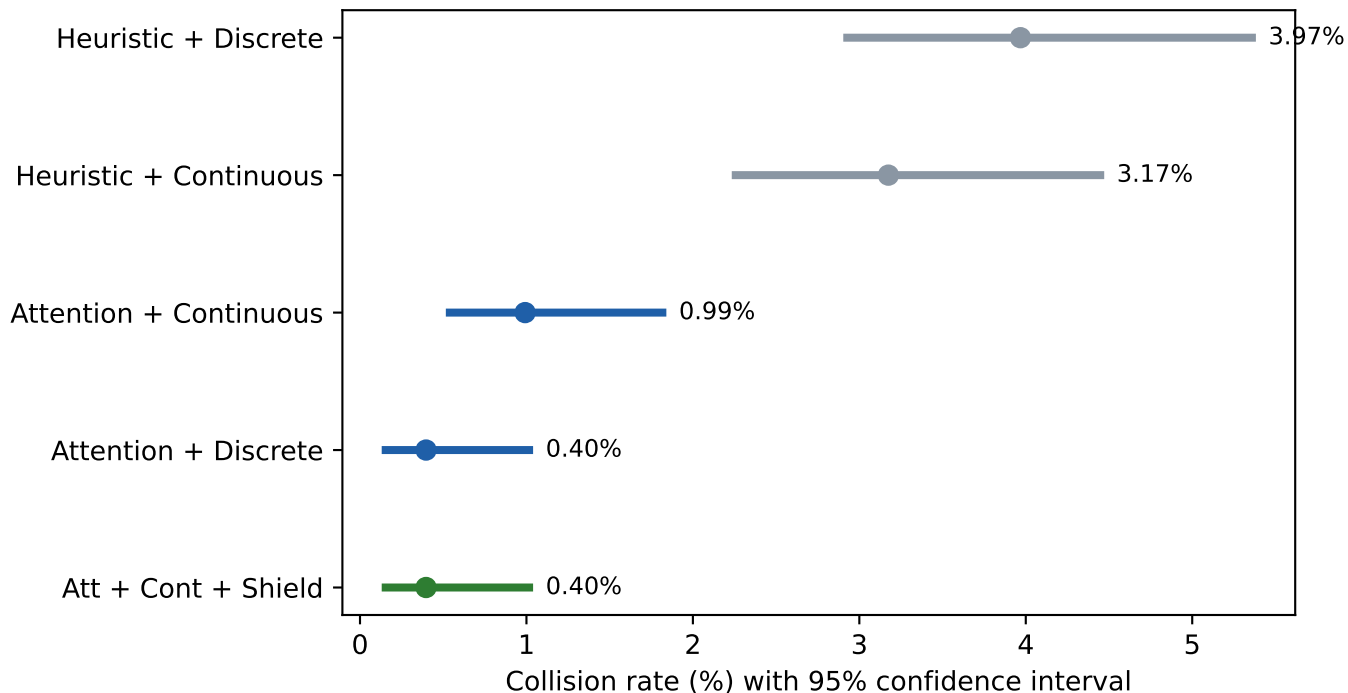
Collision rate (%) averaged over intention settings. Top: values reported in the pre-defence report. Bottom: this work on the same evaluation benchmark.

Pre-defence benchmark: reported baselines and this work



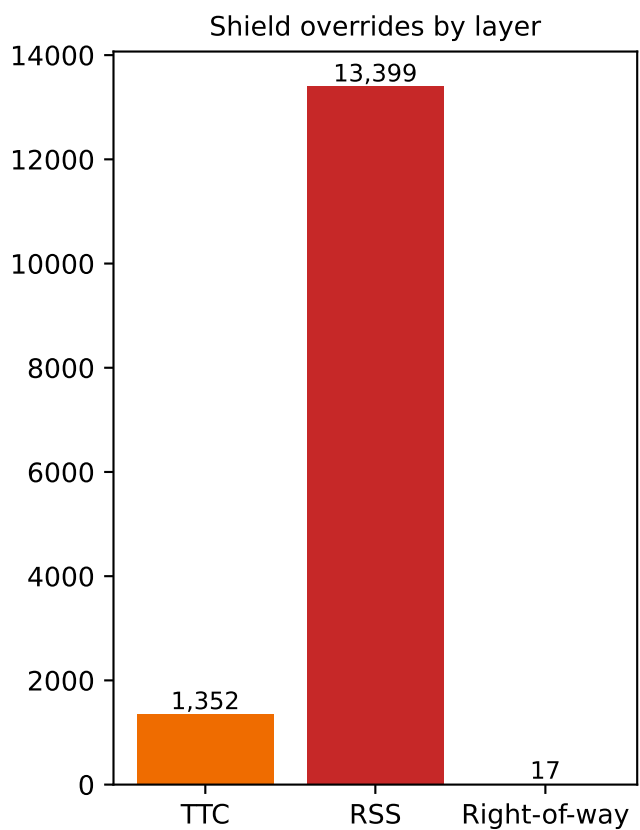
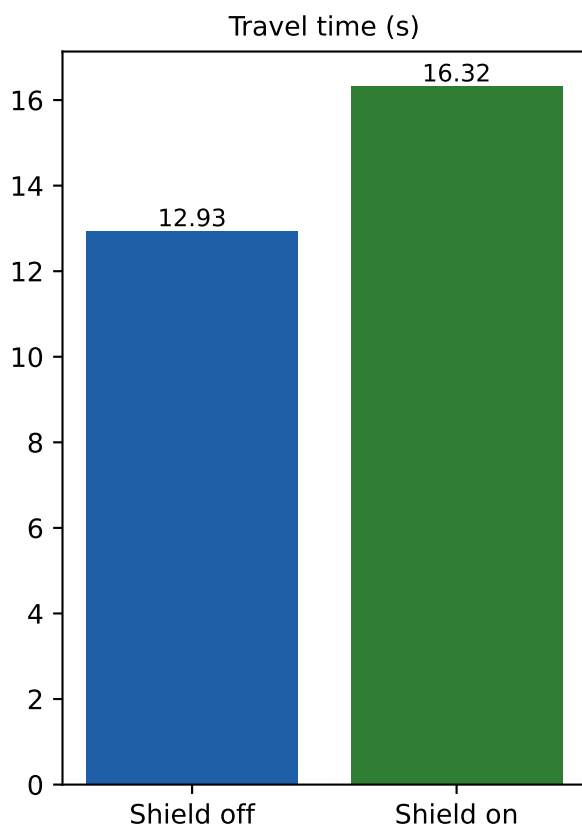
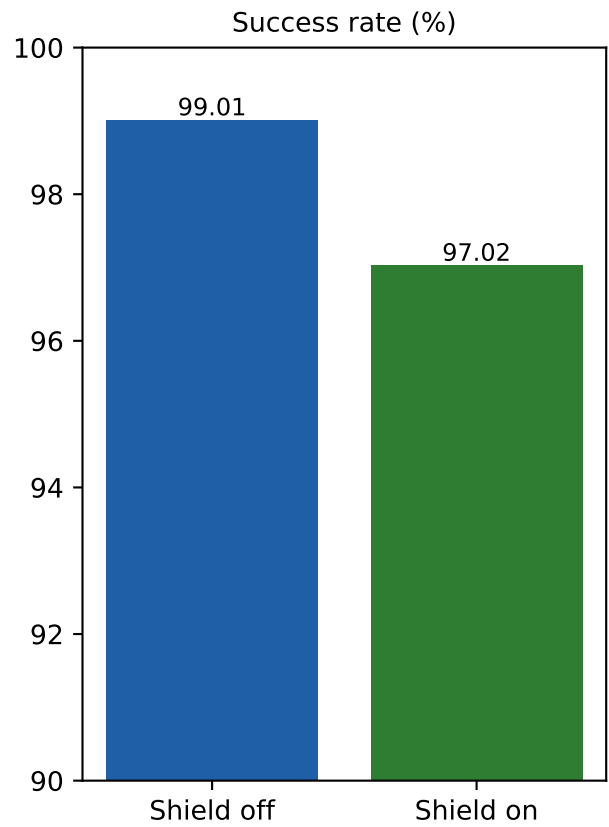
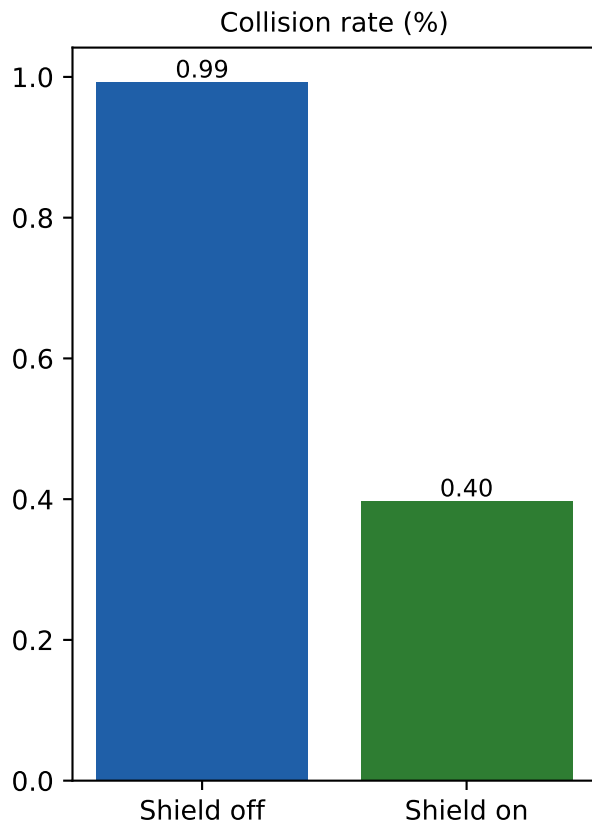
Grey: pre-defence report. Green: shielded controller from this work, trained in the revised pipeline. The 0.40% vs 0.88% difference is not statistically significant.

Two tiers: heuristic vs attention-based controllers



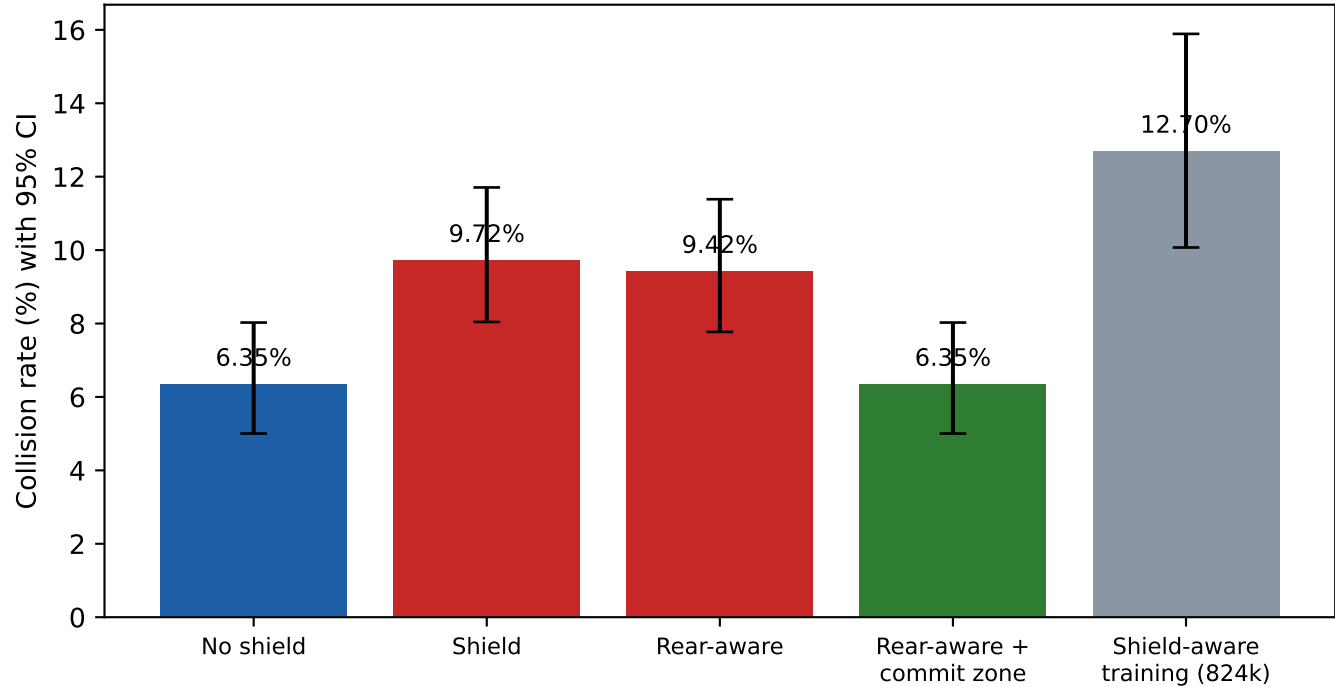
Revised pipeline, 1,008 episodes per controller. Heuristic controllers are significantly worse ($p < 0.001$); the attention-based controllers overlap and are not distinguishable.

Same policy weights, shield off vs on (benchmark)



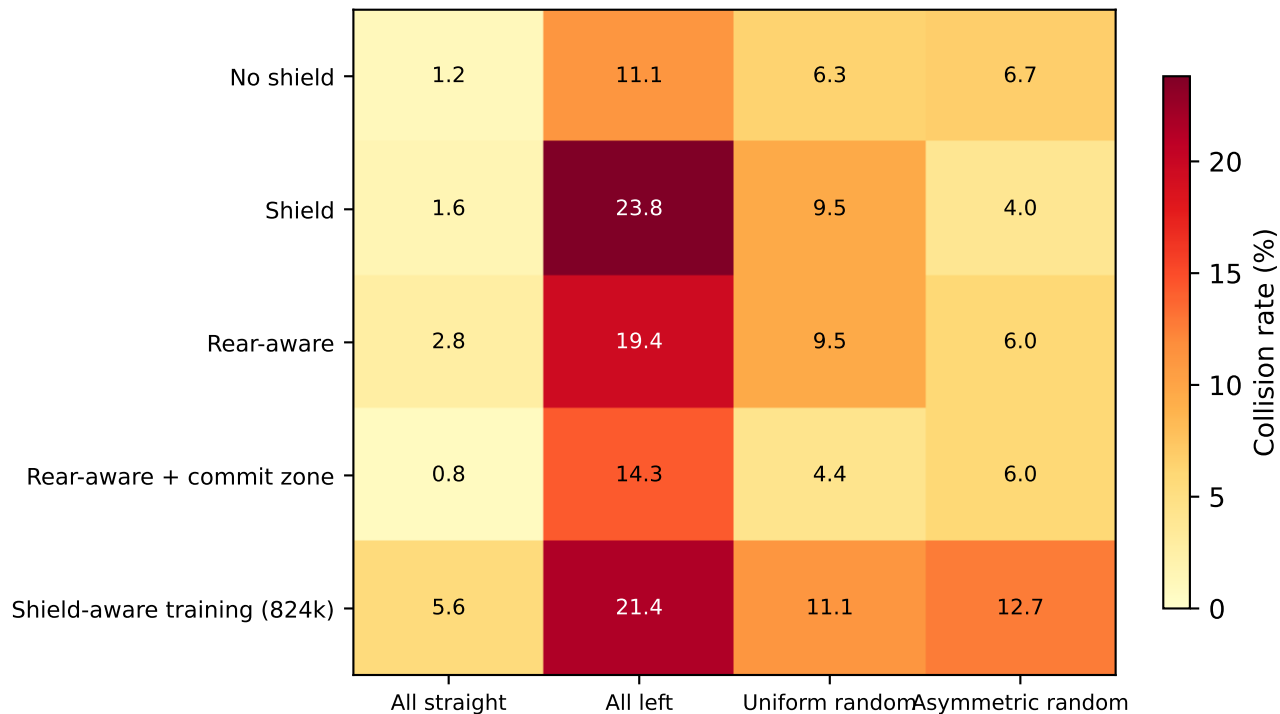
Collisions 10 vs 4 of 1,008 ($p = 0.18$). The RSS layer produces most interventions; the right-of-way layer rarely fires.

Pre-defence environment: training and evaluation



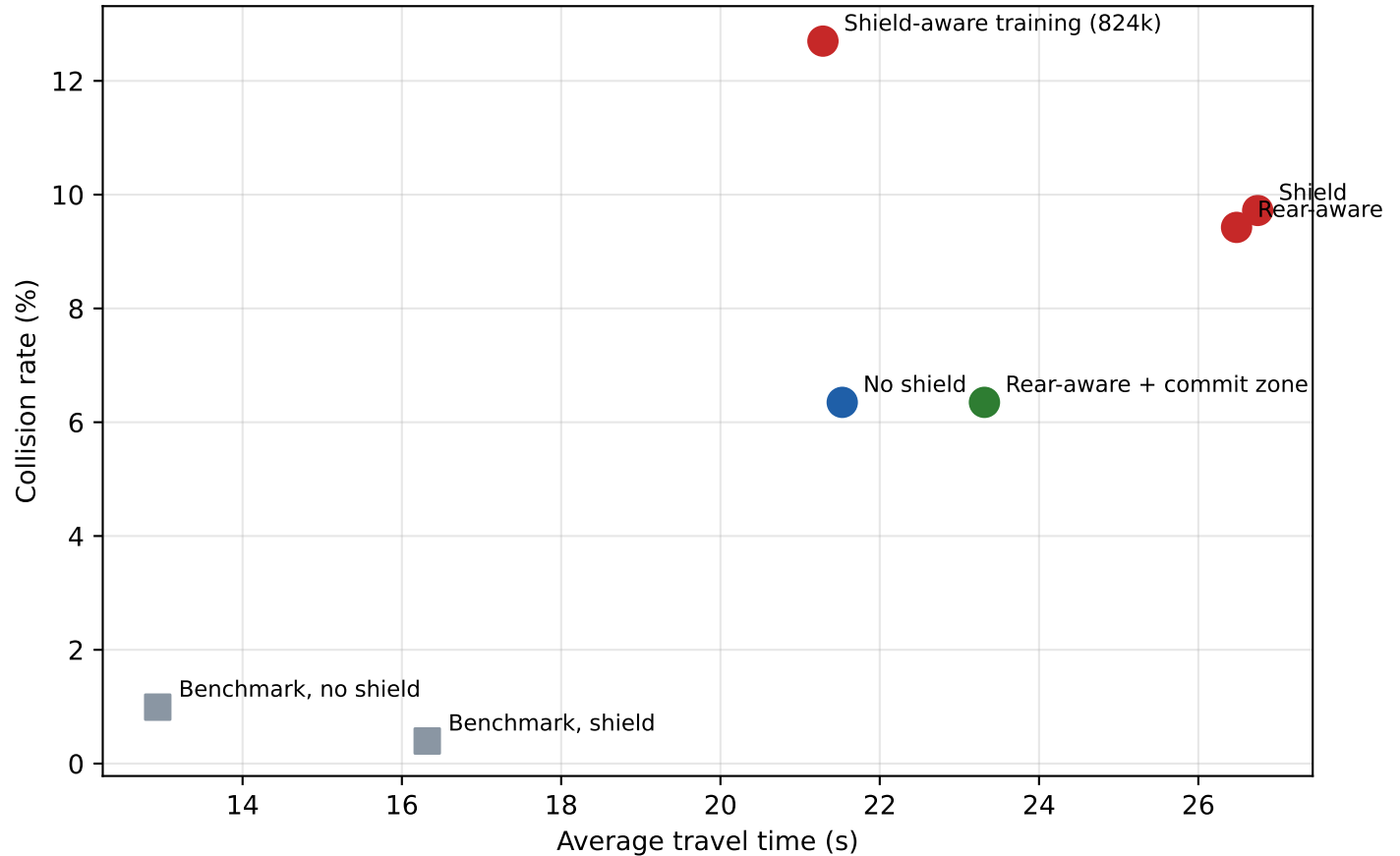
No shield: 64/1008; Shield: 98/1008, $p = 0.007$ vs no shield; Rear-aware: 95/1008, $p = 0.013$ vs no shield; Rear-aware + commit zone: 64/1008, $p = 1.000$ vs no shield; Shield-aware training (824k): 64/504, $p = 0.000$ vs no shield. Shield-aware training was stopped at 824k steps against 1.5M.

Collision rate (%) by intention, pre-defence environment



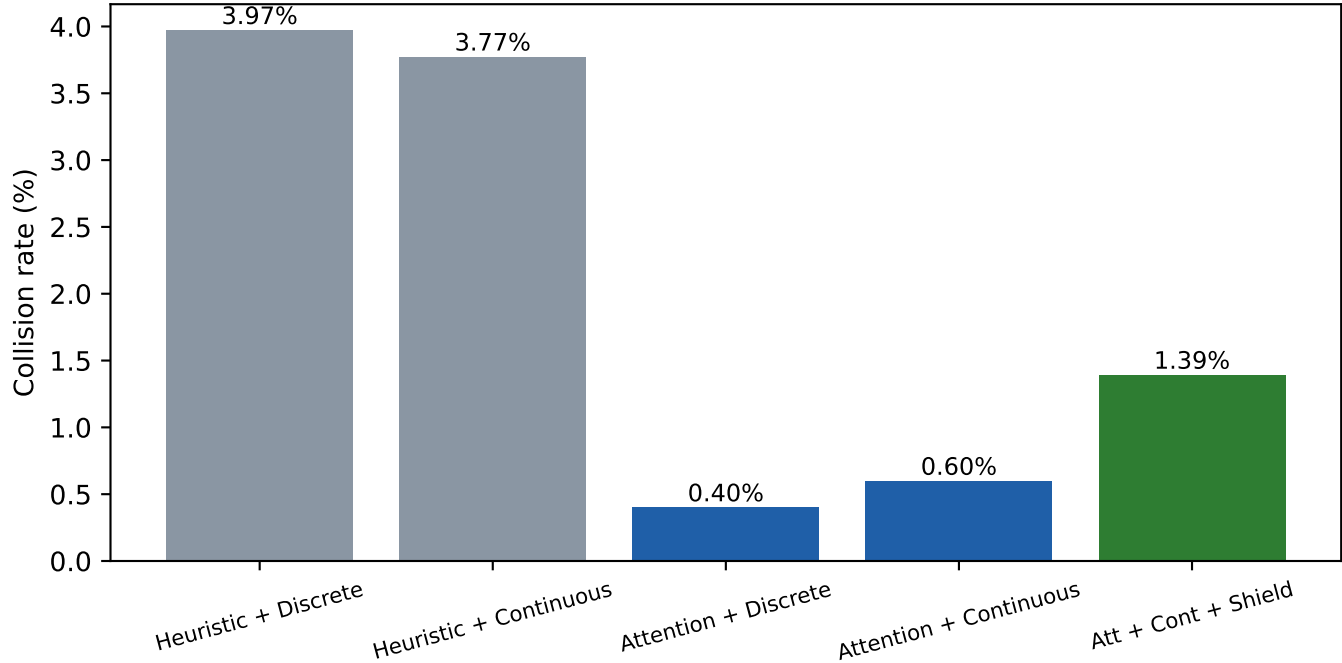
Left turns are hardest and deteriorate most under shield braking, because the agent is struck from behind by background vehicles that do not brake.

Safety vs efficiency



Lower-left is better. Circles: pre-defence environment. Squares: benchmark. Shield variants move right (slower); only the rear-aware + commit-zone variant avoids moving up (more collisions).

High-flow traffic (550 / 700 / 850 veh/h)



Higher nominal flow did not raise collision rates, most likely because congested approaches limit how many vehicles SUMO inserts. The shield did not reduce collisions here.

Limitations and future work

Limitations

Sample size: at collision rates below 1%, differences of a few collisions cannot be resolved with 1,008 episodes.

Training environment: benchmark results use a revised pipeline whose training environment is easier than the pre-defence one. The pre-defence environment results remove this difference for Attention + Continuous only.

Shield variants in the pre-defence environment (rear-aware, commit zone) were designed after the original shield's failures were observed. All variants run are reported.

Shield-aware training was stopped at 824k steps against 1.5M for the unshielded model, and no matched unshielded checkpoint was available.

Each controller was trained with a single seed.

Future work

Full-length shield-aware training against a matched baseline; calibrating the RSS trigger distance to the perception radius; multiple seeds and larger evaluation budgets; stress scenarios that raise conflict density rather than nominal inflow.

Reproducibility

All numbers are generated from eval_results/ by scripts/make_report.py and scripts/make_pdf.py. Setup and every option are documented in SETUP.md.